AI-Assisted Fact Checking Tool to Combat Online Misinformation

CS410 – Feasibility Presentation

Presented by:

Group Blue

About the Team

Johnovon Richards
- Senior computer science student at ODU
  Interested in data science and game development.

Patrick Cunanan
- Senior CS student @ ODU
  Mostly interested in web development

Shaavon Farrow
- Senior CS Student @ ODU

Anthony Oliphant
- (Bio goes here)

Colby Scroggins
- Senior CS Student at ODU

Mohumed Shemson
- (Bio goes here)

Johnovon Richards
Senior

Patrick Cunanan
Senior

Shaavon Farrow
Title

Anthony Oliphant
Title

Colby Scroggins
Senior

Mohumed Shemson
Title

Elevator Pitch

Online misinformation spreads rapidly across social media, news sites, and online communities, making it difficult for users to decipher what information they can trust, especially with the rise of AI-generated content. Existing fact-checking organizations can be limited due to recent AI innovations and usually can only verify highly viral content or posts on specific platforms.

What our project proposes is an AI-assisted fact-checking tool that allows users to verify information quickly by analyzing text, articles, or social media posts across the internet. The system will evaluate claims, search reliable sources, and provide a credibility assessment with supporting evidence.

This tool will help users quickly identify misleading information and influence them to make more informed decisions based on the content they consume.

Table of Contents

1 Societal Problem

2 Current Process

3 Proposed Software Solution

4 Features and Scope

5 Competition Analysis

6 Development Tools

7 System Architecture

8 Risks

9 Conclusion

The Misinformation Problem

The Core Issue

Internet users are constantly subjected to rising levels of misinformation stemming from online sources and platforms.

The AI Factor

The proliferation of AI-generated posts, articles, and social media accounts only worsens the problem.

Verification Gaps

While some online platforms have taken steps to implement fact-checking features, they are often platform-specific and limited by post engagement..

The Bottom Line

Users currently lack a universal, accessible, and on-demand method for quickly verifying information that they encounter on the internet.

Anatomy of the Misinformation Problem

The Scale

The spread of misinformation is not isolated. Rather, it is deeply embedded in everyday web browsing and regular social media use.

Current Solutions Fall Short

Limiting factors of current fact-checking solutions:

- Platform Restrictions - Verification tools are limited to their specific platform
- Popularity Bias - most verification only occurs on highly viewed posts, leaving the majority unverified

The User Burden

Manually validating claims requires users to conduct online searches, wade through vast amounts of information, and cross-reference sources. This can be time consuming and confusing, leading most users to skip the process entirely.

Content Generation

Creating vast amounts of fake content is easier than ever due to the proliferation of generative AI tools. Manual fact checking is easily outpaced by the scale at which content is being generated.

Current Process Flow

Encountering Online Misinformation

User encounters potentially misleading content

- User relies on platform-specific fact-checking tools
  - The post is viral enough to trigger content verification
  - The post is too new/niche to trigger content verification
- The platform requires manual fact checking from the user
  - User conducts web searches, cross-references to verify information
  - User does not take the time to verify information

- **The Core Failure**
  Currently, fact checking is limited by platform restrictions or by the user's finite time and energy. There is no universal, easy-to-use solution.

Current Process Flow

Addressing Online Misinformation

User encounters potentially misleading content

- User relies on platform-specific fact-checking tools
  - The post is viral enough to trigger content verification
  - The post is too new/niche to trigger content verification
- The platform requires manual fact checking from the user
  - User conducts web searches, cross-references to verify information
  - User does not take the time to verify information

- **The Core Failure**
  Currently, fact checking is limited by platform restrictions or by the user's finite time and energy. There is no universal, easy-to-use solution.

Expectations of the Software

What it WILL do

- Accept user input such as: links to articles, social media posts, and copied and pasted text
- Use AI models and web searches to analyze claims within the content
- Compare claims against trusted sources and databases
- Provide a credibility assessment (e.g., likely true, misleading, false, or uncertain)
- Display supporting evidence and source references used in the evaluation
- Be accessible through a browser extension or mobile application for quick fact checking

What it WILL NOT do

- Guarantee 100% accuracy in credibility assessments
- Automatically remove or censor online content
- Track users or store personal browsing data long-term
- Provide political or ideological opinions
- Replace human/user fact-checking

The Risks (Security)

Data Privacy and User Input Security

If users paste articles, social media posts, or private messages into the tool for verification, that information could contain sensitive or personal data. If the system does not properly secure stored inputs or transmissions, attackers could access this data through breaches or interception. Strong encryption and secure data handling practices would be necessary to protect user privacy.

Model Manipulation

Attackers may try to exploit or manipulate the AI system itself. For example, malicious users could attempt adversarial attacks by submitting specially crafted inputs designed to confuse the model and produce incorrect credibility assessments. Attackers could also try to probe the system repeatedly to learn how it works and then exploit weaknesses. This could reduce the reliability of the tool and allow misinformation to pass through the system undetected.

The Risks

False Positives or Incorrect Assessments

AI models rely on training data and patterns, which means they can misinterpret context, sarcasm, or complex claims. If the system gives incorrect credibility assessments, users might trust inaccurate results and spread misinformation further instead of preventing it.

Bias in Training Data and Sources

If the system mainly pulls from certain news outlets, databases, or political viewpoints, it may unintentionally reflect those biases in its credibility assessments. This could lead to some information being unfairly labeled as unreliable while other biased sources are treated as trustworthy.

Gaming of The System

If the AI relies on quantity of sources instead of true reliability, it could be tricked into validating false claims.

The Over Reliance on Automated Fact Checking

Users may rely too heavily on the AI's decision instead of thinking critically about information themselves. If people blindly trust the AI output, they may stop questioning information or verifying sources on their own.

Monetization Plans

Potential Monetization Strategies

1

Premium Tier

This could include:

- Faster analysis
- Deeper claim breakdown
- Expanded source databases

2

Enterprise / API Licensing

Offer an API for organizations such as:

- News companies
- Social media platforms
- Educational institutions

3

Advertising

- Non-intrusive ads on the free version
- Ads limited to educational or research tools to maintain credibility

Conclusion

Misinformation online is becoming harder to detect, especially with the rise of AI-generated content and the speed at which information spreads across social media and news platforms. Our proposed AI-assisted fact-checking tool aims to help users quickly evaluate the credibility of online claims by analyzing text, searching reliable sources, and providing evidence-based credibility assessments.

While there are risks such as AI bias, incorrect classifications, and potential manipulation, this system has the potential to improve how people evaluate the information they encounter online. By giving users faster access to fact-checking and supporting evidence, the tool encourages more informed decision-making and promotes digital media literacy.

Overall, this project highlights how AI can be used not just to create information, but also to help people better understand and verify the information they consume.